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Class: SY - AI \& DS
Subject: Python for Visual AI

Assignment 7

An Agri-analytics company wants to collect daily vegetable prices from government portals.<https://agmarknet.gov.in/>

```
In [ ]:  
import pandas as pd  
import matplotlib.pyplot as plt  
from sklearn.linear_model import LinearRegression
```

Loading Dataset

```
In [ ]:  
df=pd.read_csv("/content/Marketwise_Price_Arrival_01-04-2026_10-10-16_PM.csv",skiprows=2  
df
```

Out[]:

	Commodity Group	Commodity	MSP (Rs./Quintal) 2026-27	Price on 30 Mar, 2026	Price on 29 Mar, 2026	Price on 28 Mar, 2026	Arrival on 30 Mar, 2026	A
0	Cereals	Bajra(Pearl Millet/Cumbu)	2775.00	2362.87	1855.71	2550.95	413.87	4
1	Cereals	Barley(Jau)	2150.00	1987.90	1931.55	1962.55	1777.83	67
2	Cereals	Jowar(Sorghum)	3699.00	4637.66	2374.25	4132.76	149.96	
3	Cereals	Maize	2400.00	1698.31	1688.53	1739.70	18271.32	75
4	Cereals	Paddy(Common)	2369.00	2689.67	2470.59	2969.77	9803.90	94
5	Cereals	Ragi(Finger Millet)	4886.00	4328.97	3126.82	3533.31	54.20	5
6	Cereals	Wheat	2585.00	2389.95	2297.05	2479.65	49768.73	144
7	Fibre Crops	Cotton	7710.00	7783.73	7435.72	8095.73	3520.05	33
8	Oil Seeds	Copra	12100.00	22324.47	20749.88	17375.02	69.93	
9	Oil Seeds	Groundnut	7263.00	7084.98	6694.92	7278.72	5635.54	
10	Oil Seeds	Mustard	6200.00	6570.21	6411.53	6461.71	10693.51	81
11	Oil Seeds	Niger Seed(Ramtil)	9537.00	11250.00	-	10500.00	0.30	
12	Oil Seeds	Safflower	6540.00	-	-	4643.43	-	
13	Oil Seeds	Sesamum(Sesame,Gingelly,Til)	9846.00	7823.08	7496.55	11986.87	5.20	
14	Oil Seeds	Soyabean	5328.00	5147.72	5182.65	5380.13	3640.82	7
15	Oil Seeds	Sunflower	7721.00	6624.26	6483.00	6593.24	178.40	1

	Commodity Group	Commodity	MSP (Rs./Quintal) 2026-27	Price on 30 Mar, 2026	Price on 29 Mar, 2026	Price on 28 Mar, 2026	Arrival on 30 Mar, 2026	A
16	Others	Sugarcane	285.00	331.71	300.00	300.00	7025.10	
17	Pulses	Arhar(Tur/Red Gram)(Whole)	8000.00	7632.36	7390.55	7600.05	1355.21	7
18	Pulses	Bengal Gram(Gram)(Whole)	5875.00	5386.13	5174.37	5393.93	4371.52	13
19	Pulses	Black Gram(Urd Beans)(Whole)	7800.00	8528.49	7511.45	8110.81	220.00	
20	Pulses	Green Gram(Moong)(Whole)	8768.00	8906.90	6558.14	8545.69	515.42	
21	Pulses	Lentil(Masur)(Whole)	7000.00	7792.09	6762.74	8087.63	550.88	
22	Vegetables	Onion	-	1271.47	1158.03	1212.85	7575.46	29
23	Vegetables	Potato	-	738.41	951.64	834.45	12753.84	50
24	Vegetables	Tomato	-	1284.93	1387.78	1286.17	3738.68	16

Data Preprocessing

In []:

```
df.head()
```

Out[]:

	Commodity Group	Commodity	MSP (Rs./Quintal) 2026-27	Price on 30 Mar, 2026	Price on 29 Mar, 2026	Price on 28 Mar, 2026	Arrival on 30 Mar, 2026	Arrival on 29 Mar, 2026	Arrival on 28 Mar, 2026
0	Cereals	Bajra(Pearl Millet/Cumbu)	2775.00	2362.87	1855.71	2550.95	413.87	405.00	531.85
1	Cereals	Barley(Jau)	2150.00	1987.90	1931.55	1962.55	1777.83	6739.82	1473.69
2	Cereals	Jowar(Sorghum)	3699.00	4637.66	2374.25	4132.76	149.96	48.00	186.90
3	Cereals	Maize	2400.00	1698.31	1688.53	1739.70	18271.32	7520.58	12325.91
4	Cereals	Paddy(Common)	2369.00	2689.67	2470.59	2969.77	9803.90	9443.77	14498.04

In []:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 25 entries, 0 to 24
```

```
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
0	Commodity Group	25 non-null	object
1	Commodity	25 non-null	object
2	MSP (Rs./Quintal) 2026-27	25 non-null	object
3	Price on 30 Mar, 2026	25 non-null	object
4	Price on 29 Mar, 2026	25 non-null	object
5	Price on 28 Mar, 2026	25 non-null	float64
6	Arrival on 30 Mar, 2026	25 non-null	object

```
7  Arrival on 29 Mar, 2026    25 non-null    object
8  Arrival on 28 Mar, 2026    25 non-null    float64
dtypes: float64(2), object(7)
memory usage: 1.9+ KB
```

In []:

```
df.isnull().sum()
```

Out[]:

	0
Commodity Group	0
Commodity	0
MSP (Rs./Quintal) 2026-27	0
Price on 30 Mar, 2026	0
Price on 29 Mar, 2026	0
Price on 28 Mar, 2026	0
Arrival on 30 Mar, 2026	0
Arrival on 29 Mar, 2026	0
Arrival on 28 Mar, 2026	0

dtype: int64

Renaming Columns

In []:

```
df.rename(columns={
    'msp_(rs./quintal)_2026-27': 'msp',
    'price_on_30_mar,_2026': 'price_30',
    'price_on_29_mar,_2026': 'price_29',
    'price_on_28_mar,_2026': 'price_28',
    'arrival_on_30_mar,_2026': 'arrival_30',
    'arrival_on_29_mar,_2026': 'arrival_29',
    'arrival_on_28_mar,_2026': 'arrival_28'
}, inplace=True)
```

Convert categorical columns into numeric

In []:

```
cols = [
    'MSP (Rs./Quintal) 2026-27',
    'Price on 30 Mar, 2026',
    'Price on 29 Mar, 2026',
    'Price on 28 Mar, 2026',
    'Arrival on 30 Mar, 2026',
    'Arrival on 29 Mar, 2026',
    'Arrival on 28 Mar, 2026'
]
```

```
for col in cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

MSP Comparison

The market price is compared with MSP to determine farmer profit.

$$Above\ MSP = \begin{cases} 1 & \text{if Price} > \text{MSP} \\ 0 & \text{otherwise} \end{cases}$$

In []:

```
df['above_msp_30'] = df['Price on 30 Mar, 2026'] > df['MSP (Rs./Quintal) 2026-27']
df['above_msp_29'] = df['Price on 29 Mar, 2026'] > df['MSP (Rs./Quintal) 2026-27']
df['above_msp_28'] = df['Price on 28 Mar, 2026'] > df['MSP (Rs./Quintal) 2026-27']
```

Average Price Calculation

Average price is calculated to understand overall trends.

$$AveragePrice = \frac{\sum_{i=1}^n Price_i}{n}$$

In []:

```
print("Avg Price 30 March:\n", df['Price on 30 Mar, 2026'].mean())
print("Avg Price 29 March:\n", df['Price on 29 Mar, 2026'].mean())
print("Avg Price 28 March:\n", df['Price on 28 Mar, 2026'].mean())

print(df.groupby('Commodity')[['Price on 30 Mar, 2026', 'Price on 29 Mar, 2026', 'Price on
```

Avg Price 30 March:

5690.6779166666665

Avg Price 29 March:

4930.15

Avg Price 28 March:

5562.204800000001

Commodity	Price on 30 Mar, 2026	Price on 29 Mar, 2026	\
Arhar(Tur/Red Gram)(Whole)	7632.36	7390.55	
Bajra(Pearl Millet/Cumbu)	2362.87	1855.71	
Barley(Jau)	1987.90	1931.55	
Bengal Gram(Gram)(Whole)	5386.13	5174.37	
Black Gram(Urd Beans)(Whole)	8528.49	7511.45	
Copra	22324.47	20749.88	
Cotton	7783.73	7435.72	
Green Gram(Moong)(Whole)	8906.90	6558.14	
Groundnut	7084.98	6694.92	
Jowar(Sorghum)	4637.66	2374.25	
Lentil(Masur)(Whole)	7792.09	6762.74	
Maize	1698.31	1688.53	
Mustard	6570.21	6411.53	
Niger Seed(Ramtil)	11250.00	NaN	
Onion	1271.47	1158.03	
Paddy(Common)	2689.67	2470.59	
Potato	738.41	951.64	
Ragi(Finger Millet)	4328.97	3126.82	

Safflower	NaN	NaN
Sesamum(Sesame,Gingelly,Til)	7823.08	7496.55
Soyabean	5147.72	5182.65
Sugarcane	331.71	300.00
Sunflower	6624.26	6483.00
Tomato	1284.93	1387.78
Wheat	2389.95	2297.05

Price on 28 Mar, 2026

Commodity	
Arhar(Tur/Red Gram)(Whole)	7600.05
Bajra(Pearl Millet/Cumbu)	2550.95
Barley(Jau)	1962.55
Bengal Gram(Gram)(Whole)	5393.93
Black Gram(Urd Beans)(Whole)	8110.81
Copra	17375.02
Cotton	8095.73
Green Gram(Moong)(Whole)	8545.69
Groundnut	7278.72
Jowar(Sorghum)	4132.76
Lentil(Masur)(Whole)	8087.63
Maize	1739.70
Mustard	6461.71
Niger Seed(Ramtil)	10500.00
Onion	1212.85
Paddy(Common)	2969.77
Potato	834.45
Ragi(Finger Millet)	3533.31
Safflower	4643.43
Sesamum(Sesame,Gingelly,Til)	11986.87
Soyabean	5380.13
Sugarcane	300.00
Sunflower	6593.24
Tomato	1286.17
Wheat	2479.65

Group-wise Analysis

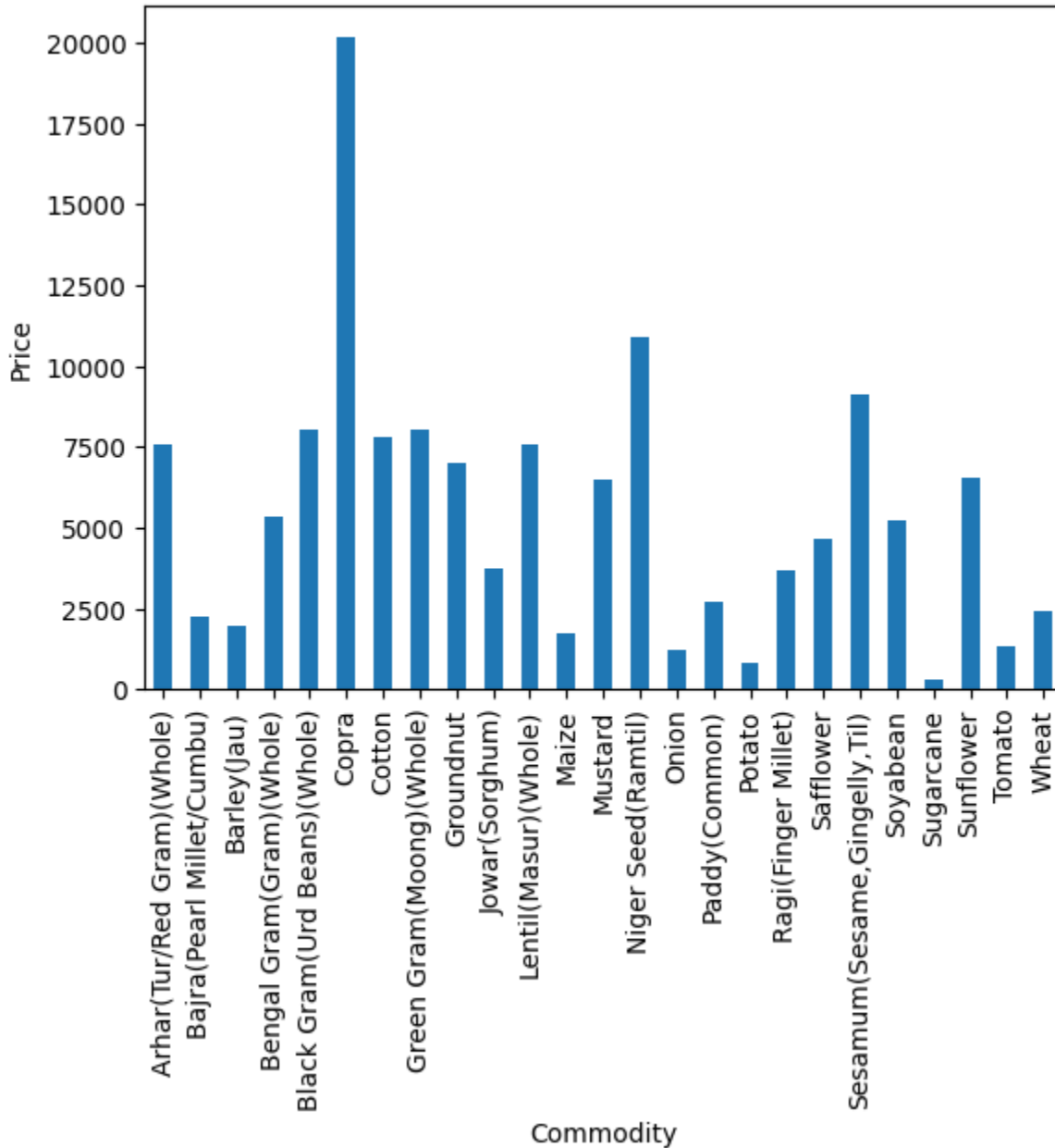
Prices are grouped by commodity.

$$\bar{P}_{commodity} = \frac{\sum Price}{Count}$$

In []:

```
df_final.groupby('Commodity')['price'].mean().plot(kind='bar')
plt.title("Average Prices of Vegetables")
plt.xlabel("Commodity")
plt.ylabel("Price")
plt.show()
```

Average Prices of Vegetables



Price Trend Analysis

Price trend shows variation over time.

$$Trend = f(\text{Time})$$

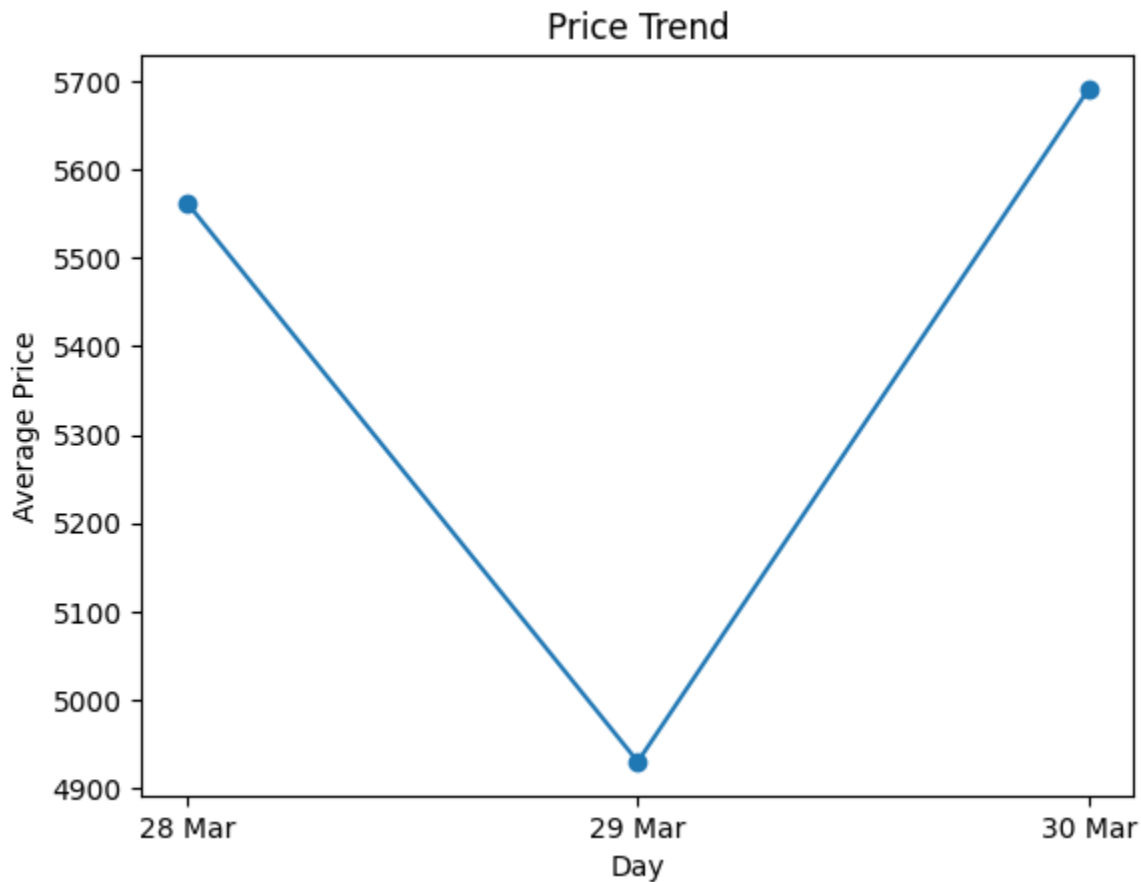
In []:

```
import matplotlib.pyplot as plt

avg_prices = [
    df['Price on 28 Mar, 2026'].mean(),
    df['Price on 29 Mar, 2026'].mean(),
    df['Price on 30 Mar, 2026'].mean()
]

days = ['28 Mar', '29 Mar', '30 Mar']
```

```
plt.plot(days, avg_prices, marker='o')
plt.title("Price Trend")
plt.xlabel("Day")
plt.ylabel("Average Price")
plt.show()
```



Linear Regression Model

Linear regression is used for price prediction.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Model Explanation

The linear regression model is given by:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

where,

y = Predicted Price

x_1 = MSP (Minimum Support Price)

x_2 = Price on previous day

x_3 = Arrival of commodity

$$\beta_0, \beta_1, \beta_2, \beta_3 = \text{Model coefficients}$$

Prediction

The model predicts future prices.

$$\hat{y} = f(X)$$

In []:

```
X = df[['MSP (Rs./Quintal) 2026-27', 'Price on 28 Mar, 2026', 'Price on 29 Mar, 2026', 'Price on 30 Mar, 2026']
y = df['Price on 30 Mar, 2026']

valid_indices = X.dropna().index
X = X.loc[valid_indices]
y = y.loc[valid_indices]

model = LinearRegression()
model.fit(X, y)

print(model.predict([[2500, 2000, 2100, 500]]))

[2545.07781724]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(
```

Saving File

In []:

```
df.to_csv("final_dataset.csv", index=False)
```

Conclusion

The experiment successfully analyzed vegetable price data from the AGMARKNET dataset to understand market trends. It compared daily prices with MSP to evaluate farmer profitability. The relationship between arrivals and prices was studied to identify supply-demand effects. A linear regression model was applied to predict future prices, supporting data-driven agricultural decisions.

In []: